Assignment No. 4

Quick Work
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Date:
M T W T F S S

Ass-1

Q1] Print an nxn star square

Time Complexity - $O(n^2)$
Space Complexity - $O(1)$
There is inner for loop
All variable were constant

Q2] Print the following patter.

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

Time Complexity $O(n^2)$
Space Complexity - $O(1)$

Q3] Print the following pattern

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

TC = $O(n^2)$
= $O(1)$

Ass 2

Q1] Hollow Square

```
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```

TC - $O(n^2)$
SC - $O(1)$

Q.2) Diagonal Square (x pattern)

```
*     *
  *   *
    *
  *   *
*     *
```

$T(c) - O(n^2)$

$S(c) - O(1)$

Q.3) Right Traingle (Derived from Square).

```
*
*  *
*  *  *
*  *  *  *
*  *  *  *  *
```

$T(c) - O(n^2)$

$S(c) - O(1)$

Q.4) Inverted Right Traingle.

```
*  *  *  *  *
*  *  *  *
*  *  *
*  *
*
```

$T(c) - O(n^2)$

$S(c) - O(1)$

Q.5) ChessBoard Pattern

```
*  #  *  #
#  *  #  *
*  #  *  #
#  *  #  *
```

$T(c) - O(n^2)$

$S(c) - O(1)$

Q1) Double number Pyramid

```
        1
      1 1 2
    1 2   3
  1 2     3 4
    1 2   3
      1 2
        1
```

$T(c) = O(n^2) + O(n) + O(n^2) + O(n)$
$= 2(2n^2) + 2O(n)^2$
$= O(n^2)$

$SC = O(1)$

Q2) Number Butterfly

```
1                 1
1 2               2 1
1 2 3           3 2 1
1 2 3 4 4 3 2 1
1 2 3 4 4 3 2 1
1 2 3           3 2 1
1 2               2 1
1                 1
```

$T(c) = O(n) + O(n^2) + O(n)$
$\rightarrow O(n^2)$

$SC = O(1)$

Q3)

```
1       1
  2   2
    3
  4   4
5       5
```

$T(c) = O(n^2)$
$S(c) = O(1)$

Q.4)

```
3 3 3 3 3
3 2 2 2 3
3 2 1 2 3
3 2 2 2 3
3 3 3 3 3
```

$T(n) - O(n^2)$

$S(n) - O(1)$

Q.5)

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

$T(n) - O(n^2)$

$S(n) - O(n^2)$

Q.6)

```
      1
    2   3
  4   5   6
7   8   9   10
  11  12  13
    14  15
      16
```

$T(n) - O(n^2)$

$S(n) - O(n)$

Q.7) Binary Diamond

```
      1
    0   1
  1   0   1
0   1   0   1
  1   0   1
    0   1
      1
```

$T(n) - O(n^2$

$S(n) - O(1)$